



NORMANHURST BOYS HIGH SCHOOL
NEW SOUTH WALES

2015
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

BOSTES Number: _____

CLASS (Please circle): 12M1 12M2 12M3 12M4 12M5

Mathematics Extension 1

General Instructions

- Reading time - 5 minutes
- Working time - 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14, show relevant mathematical reasoning and/or calculations
- Begin each question on a separate writing booklet

Total marks - 70

Section I

Pages 2-4

10 marks

- Attempt Questions 1-10
- Answer on the Multiple Choice answer sheet provided
- Allow about 15 minutes for this section

Section II

Pages 5-9

60 marks

- Attempt Questions 11-14
- Allow about 1 hour 45 minutes for this section

Students are advised that this is a school-based examination only and cannot in any way guarantee the content or format of future Higher School Certificate Examinations.

Section I

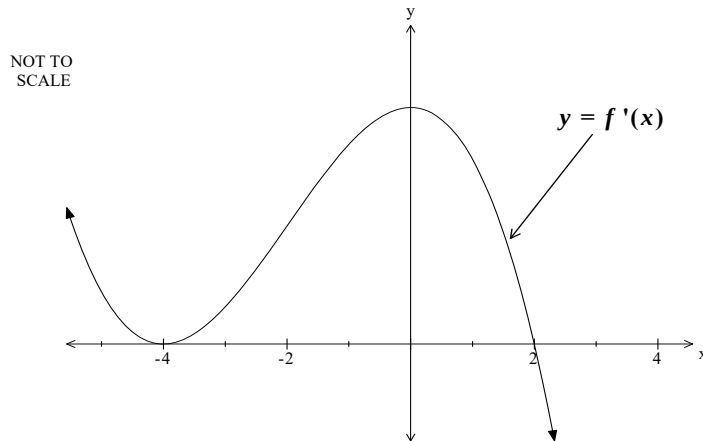
10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1



The diagram above represents a sketch of the gradient function of the curve $y = f(x)$. Which of the following is a true statement? The curve $y = f(x)$ has

- (A) a minimum turning point occurs at $x = -4$
- (B) a horizontal point of inflexion occurs at $x = 2$
- (C) a horizontal point of inflexion occurs at $x = -4$
- (D) a maximum turning point occurs at $x = 0$.

2 Solve $\frac{3}{x(2x-1)} \geq 1$

- (A) $x \leq -1$, $0 < x < \frac{1}{2}$ and $x \geq \frac{3}{2}$
- (B) $-1 \leq x < 0$ and $\frac{1}{2} < x \leq \frac{3}{2}$
- (C) $-\frac{3}{2} \leq x \leq -\frac{1}{2}$ and $0 < x \leq 1$
- (D) $x \leq -\frac{3}{2}$, $-\frac{1}{2} \leq x < 0$ and $x \geq 1$

3 If $\sin x = \frac{3}{5}$, $\frac{\pi}{2} \leq x \leq \pi$, evaluate $\tan 2x$.

(A) $-\frac{7}{24}$

(B) $-\frac{24}{7}$

(C) $\frac{7}{24}$

(D) $\frac{24}{7}$

4 Find the acute angle between the tangents to the graphs $y = x$ and $y = x^3$ at the point (1,1).

(A) 27°

(B) 30°

(C) 45°

(D) 63°

5 The polynomial $P(x) = 2x^3 + kx^2 + 5x - 2$ has $(x + 2)$ as a factor. Find the value of k .

(A) -7

(B) 7

(C) -12

(D) 12

6 Find $\frac{d}{dx}(\sqrt{1-x^2} + x \sin^{-1} x)$

(A) $\frac{-2}{\sqrt{1-x^2}}$

(B) $\frac{-1}{\sqrt{1-x^2}}$

(C) $\cos^{-1} x$

(D) $\sin^{-1} x$

- 7 Find the domain and range of $y = 3 \cos^{-1} \pi x$
- (A) Domain: $0 \leq x \leq 3$ Range: $0 \leq y \leq \pi$
- (B) Domain: $-1 \leq x \leq 1$ Range: $0 \leq y \leq \pi$
- (C) Domain: $-\pi \leq x \leq \pi$ Range: $0 \leq y \leq 3\pi$
- (D) Domain: $-\frac{1}{\pi} \leq x \leq \frac{1}{\pi}$ Range: $0 \leq y \leq 3\pi$
- 8 A stone is thrown at an angle of α to the horizontal. The position of the stone at time t seconds is given by $x = Vt \cos \alpha$ and $y = Vt \sin \alpha - \frac{1}{2}gt^2$ where $g \text{ m/s}^2$ is the acceleration due to gravity and $v \text{ m/s}$ is the initial velocity of projection.
- What is the maximum height reached by the stone?
- (A) $\frac{V \sin \alpha}{g}$
- (B) $\frac{g \sin \alpha}{V}$
- (C) $\frac{V^2 \sin^2 \alpha}{2g}$
- (D) $\frac{g \sin^2 \alpha}{2V^2}$
- 9 The volume of a sphere of radius 8 mm is increasing at a constant rate of 50 mm^3 .
- Determine the rate of increase of the surface area of the sphere.
- (A) $0.06 \text{ mm}^2/\text{s}$
- (B) $0.60 \text{ mm}^2/\text{s}$
- (C) $1.25 \text{ mm}^2/\text{s}$
- (D) $12.50 \text{ mm}^2/\text{s}$
- 10 The acceleration of a particle is defined in terms of its position by $\ddot{x} = 2x^3 + 4x$. The particle is initially 2m to the right of the origin, travelling with velocity 6ms^{-1} . Find the minimum speed of the particle.
- (A) 6 ms^{-1}
- (B) 16 ms^{-1}
- (C) 20 ms^{-1}
- (D) 36 ms^{-1}

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	Use a SEPARATE writing booklet.	Marks
(a) (i)	Evaluate $\int_1^2 \frac{dx}{x}$.	1
(ii)	Use Simpson's rule with 3 function values to approximate $\int_1^2 \frac{dx}{x}$.	1
(iii)	Use your results to parts (i) and (ii) to obtain an approximation for e . Give your answer correct to 3 decimal places.	2
(b)	Using one application of Newton's method with $x = \frac{\pi}{2}$ as the first approximation, find the second approximation to the root of the equation $\cos x = 2x - 3$. Correct answer to 3 decimal places.	2
(c)	The polynomial $P(x) = x^3 + bx^2 + cx + d$ has roots 0, 3 and -3 .	
(i)	What are the values of b , c and d ?	2
(ii)	Without using calculus, sketch the graph of $y = P(x)$.	1
(iii)	Hence or otherwise, solve the inequality $\frac{x^2 - 9}{x} > 0$	1
(d)	Consider the function $f(x) = \tan^{-1} \frac{1}{x}$. Find the slope of the tangent to the curve where the function $y = f(x)$ cuts the y-axis.	2
(e)	Find the exact value of $\int_{-1}^1 \sqrt{4 - x^2} dx$, using the substitution $x = 2 \sin \theta$.	3

Question 12 (15 marks) Use a SEPARATE writing booklet. **Marks**

- (a) The point $C(-28, 19)$ divides the interval AB externally in the ratio $k: 5$. **2**
Find the value of k if A is the point $(-4, 3)$ and B is the point $(2, -1)$

- (b) (i) Show that $\sin\left(x + \frac{\pi}{4}\right) = \frac{\sin x + \cos x}{\sqrt{2}}$ **1**

- (ii) Hence or otherwise, solve $\frac{\sin x + \cos x}{\sqrt{2}} = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$. **2**

- (c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

- (i) Show that the equation of the normal to the parabola at P is given by **1**
 $x + py = 2ap + ap^3$.

- (ii) Find the co-ordinates of R , the point of intersection of the normals at P and Q , in terms of p and q . **2**

- (iii) If $pq = -2$, find the cartesian equation of the locus of R . **2**

- (d) Evaluate $\int_0^1 x^2 \sqrt{1 + 3x^3} dx$ using the substitution $u = 1 + 3x^3$. **2**

- (e) After t years the number of animals, N , in a national park decreases according to the equation:

$$\frac{dN}{dt} = -0.09(N - 100)$$

The initial number of animals in the national park is 400.

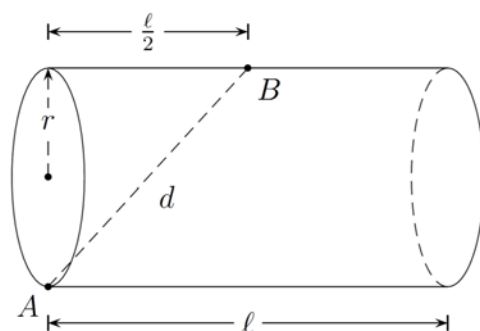
- (i) Verify that $N = 100 + Ae^{-0.09t}$ is a solution of the above equation, **1**
where A is a constant.

- (ii) How many years does it take for the number of animals to reach 150? **2**

Question 13 (15 marks) Use a SEPARATE writing booklet.

Marks

(a)

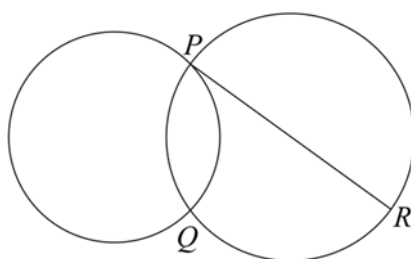


The diagram shows a cylindrical barrel of length l and radius r . The point A is at one end of the barrel, at the very bottom of the rim. The point B is at the very top of the barrel, half-way along its length. The length of AB is d .

(i) Show that the volume of the barrel is $V = \frac{\pi l}{4} \left(d^2 - \frac{l^2}{4} \right)$. 1

(ii) Find l in terms of d if the barrel has maximum volume for the given d . 2

(b) Two circles are intersecting at P and Q . The diameter of one of the circles is PR .



Copy this diagram into your writing booklet.

(i) Draw a straight line through P , parallel to QR to meet the circle PRQ at S and the other circle at T . Prove that QS is also diameter of the circle. 2

(ii) Prove that the circles have equal radii if TQ is parallel to PR . 2

Question 13 continues on page 9

Question 13 (continued)

- (c) A rocket is fired from a pontoon on the sea. The rocket is aimed at a $60m$ high cliff, $240m$ from the pontoon. The angle of projection of the rocket is 45° and its initial velocity is $40\sqrt{2}ms^{-1}$.
- (i) Taking the point of projection as the origin O , derive expressions for the horizontal component x and vertical component y of the position of the rocket at time t seconds. **2**
(Assume the acceleration due to gravity is $10ms^{-2}$)
- (ii) Show that the path of the rocket is given by the equation **1**
- $$y = x - \frac{x^2}{320}$$
- (iii) Find the time taken for the rocket to land on top of the cliff. **2**
- (iv) Find the exact velocity of the rocket when it reaches this point. **3**
(Hint: velocity includes magnitude and direction)

Question 14 (15 marks) Use a SEPARATE writing booklet. **Marks**

(a) Use mathematical induction to prove that $3^n + 7^{n+1}$ is divisible by 4 for all integers $n \geq 1$. **3**

(b) The velocity of a particle moving in a straight line is given by

$$v = 10 - x$$
where x metres is the displacement from a fixed point O and v is the velocity in metres per second. Initially the particle is at O.

(i) Show that the acceleration of the particle is given by **1**

$$\ddot{x} = x - 10$$

(ii) Express x in terms of time t . **3**

(iii) What is the limiting position of the particle? **1**

(c) A particle moves in a straight line and its displacement x metres from a fixed point O at any time t seconds is given by the equation

$$x = 4 \cos^2 t - 1.$$

(i) Prove that the particle is undergoing simple harmonic motion. **2**

(ii) State the period of the motion. **1**

(iii) Sketch the graph $x = 4 \cos^2 t - 1$ for $0 \leq t \leq \pi$. **2**
Clearly show the times when the particle passes through O.

(iv) Find the time when the velocity of the particle is increasing most rapidly for $0 \leq t \leq \pi$. **2**

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

1) (C)

$$2) \frac{3}{x(2x-1)} > 1$$

$$x^2(2x-1)^2 \times \frac{3}{x(2x-1)} > x^2(2x-1)^2$$

$$3x(2x-1) > x^2(2x-1)^2$$

$$x(2x-1)[3-x(2x-1)] > 0$$

$$x(2x-1)(2x^2-x-3) \leq 0$$

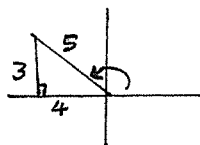
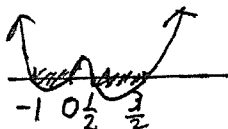
$$x(2x-1)(2x-3)(x+1) \leq 0$$

$$\therefore -1 \leq x < 0 \quad \& \quad \frac{1}{2} \leq x \leq \frac{3}{2}$$

(B)

$$3) \tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$= \frac{2(-\frac{3}{4})}{1 - (-\frac{3}{4})^2} = -\frac{24}{7}$$



(B)

$$4) y = x$$

$$y = x^3$$

$$y' = 3x^2$$

$$m_2 = 3 \quad \text{at } (1, 1)$$

$$\tan \theta = \left| \frac{1-3}{1+(1)(3)} \right| = \frac{1}{2}$$

$$\therefore \theta \doteq 27^\circ$$

(A)

$$5) p(x) = 2x^3 + kx^2 + 5x - 2$$

$$p(-2) = 0$$

$$2(-2)^3 + k(-2)^2 + 5(-2) - 2 = 0$$

$$-16 + 4k - 12 = 0$$

$$k = 7$$

(B)

$$6) \frac{d}{dx} (\sqrt{1-x^2} + x \sin^{-1} x)$$

$$= \frac{1}{2}(1-x^2)^{-\frac{1}{2}}(-2x) + x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x$$

$$= \sin^{-1} x$$

(D)

$$7) y = 3 \cos^{-1} \pi x$$

$$D: -1 < \pi x < 1$$

$$-\frac{1}{\pi} < x < \frac{1}{\pi}$$

$$R: 0 \leq y \leq 3\pi$$

(D)

$$8) y = V \sin \alpha - \frac{gt^2}{2} \quad \text{--- (1)}$$

$$\dot{y} = V \sin \alpha - gt$$

At max height, $\dot{y} = 0$

$$V \sin \alpha - gt = 0$$

$$t = \frac{V \sin \alpha}{g}$$

Sub into (1),

$$\text{Max } h = V \sin \alpha \times \frac{V \sin \alpha}{g} - \frac{g}{2} \left(\frac{V \sin \alpha}{g} \right)^2$$

$$= \frac{V^2 \sin^2 \alpha}{2g}$$

(C)

$$9) V = \frac{4}{3} \pi r^3$$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$50 = 4\pi r^2 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{25}{128\pi}$$

$$S = 4\pi r^2$$

$$\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt}$$

$$= 8\pi r \times \frac{25}{128\pi}$$

$$= 12.5 \text{ mm}^2/\text{s}$$

(D)

$$10) \ddot{x} = 2x^3 + 4x$$

$$\frac{d}{dx} \left(\frac{v^2}{2} \right) = 2x^3 + 4x$$

$$\frac{v^2}{2} = \frac{x^4}{2} + 2x^2 + C$$

$$\text{when } t=0, x=2, v=6$$

$$\therefore \frac{36}{2} = \frac{16}{2} + 8 + C$$

$$C = 2$$

$$\therefore v^2 = (x^2 + 2)^2$$

$$\text{When } t=0, x=2, v=6 > 0$$

$$\therefore v = x^2 + 2$$

Hence $v > 6$ as the particle never

changes direction and it is moving with increasing speeds.

(A)

$$11) a) i) \int_1^2 \frac{dx}{x} = [\ln x]_1^2$$

$$= \ln 2 - \ln 1$$

$$= \ln 2 \quad \checkmark$$

$$ii) \int_1^2 \frac{dx}{x} \div \frac{0.6}{3} \left(1 + 4 \times \frac{1}{1.5} + \frac{1}{2} \right)$$

$$= \frac{25}{36} \quad \checkmark$$

$$iii) \ln 2 \div \frac{25}{36}$$

$$2 \div e^{\frac{25}{36}} \quad \checkmark$$

$$e \div 2^{\frac{36}{25}}$$

$$= 2.713 \text{ (3 dec pl.)} \quad \checkmark$$

$$b) f(x) = \cos x - 2x + 3$$

$$f'(x) = -\sin x - 2$$

$$f'\left(\frac{\pi}{2}\right) = \cos \frac{\pi}{2} - 2\left(\frac{\pi}{2}\right) + 3$$

$$= 3 - \pi$$

$$f'\left(\frac{\pi}{2}\right) = -\sin \frac{\pi}{2} - 2$$

$$= -3$$

$$\therefore x_1 = \frac{\pi}{2} - \frac{3-\pi}{-3} \quad \checkmark$$

$$= \frac{\pi}{2} + 1$$

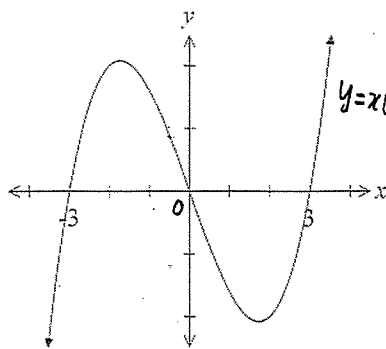
$$= 1.524 \text{ (3 dec pl.)} \quad \checkmark$$

$$c) i) x^3 + bx^2 + cx + d = x(x-3)(x+3)$$

$$= x^3 - 9x \quad \checkmark$$

$$\therefore \left. \begin{array}{l} b=0 \\ c=-9 \\ d=0 \end{array} \right\} \quad \checkmark$$

ii)



$$iii) x^2 \times \frac{x^2 - 9}{x} > 0 \times x^2$$

$$x(x^2 - 9) > 0$$

$$\text{From graph, } -3 < x < 0 \text{ or } x > 3 \quad \checkmark \quad ①$$

$$d) y = \tan^{-1} \frac{1}{x}$$

$$y' = \frac{1}{1 + \left(\frac{1}{x}\right)^2} \left(-\frac{1}{x^2}\right) \quad \checkmark$$

$$= -\frac{1}{x^2 + 1}$$

$$= -1 \text{ at } x=0 \quad \checkmark$$

①

①

②

$$e) \int_{-1}^1 \sqrt{4-x^2} dx$$

$$= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \sqrt{4-4\sin^2\theta} \times 2\cos\theta d\theta \quad \checkmark$$

$$= 4 \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2\theta d\theta$$

$$= 8 \int_0^{\frac{\pi}{6}} \frac{1}{2} (\cos 2\theta + 1) d\theta \quad \checkmark$$

$$= 4 \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\frac{\pi}{6}}$$

$$= 4 \left(\frac{1}{2} \sin \frac{\pi}{3} + \frac{\pi}{6} - 0 - 0 \right)$$

$$= 4 \left(\frac{1}{2} \times \frac{\sqrt{3}}{2} + \frac{\pi}{6} \right)$$

$$= \sqrt{3} + \frac{2\pi}{3} \quad \checkmark$$

②

③

$$12) a) -28 = \frac{2(-k) + (-4)(5)}{-k+5} \quad \checkmark$$

$$k = 4 \quad \checkmark$$

②

$$b) i) LHS = \sin\left(x + \frac{\pi}{4}\right)$$

$$= \sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4}$$

$$= (\sin x) \times \frac{1}{\sqrt{2}} + (\cos x) \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sin x + \cos x}{\sqrt{2}} \quad \checkmark$$

$$= RHS$$

①

$$ii) \sin\left(x + \frac{\pi}{2}\right) = \frac{\sqrt{3}}{2}$$

$$x + \frac{\pi}{2} = \frac{\pi}{3} \text{ or } \frac{2\pi}{3} \quad \checkmark$$

$$\therefore x = \frac{\pi}{12} \text{ or } \frac{5\pi}{12} \quad \checkmark$$

②

Discontinuity
at $x=0$ (1 mark)

12c) i)

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$= \frac{2ap}{2a}$$

$$= p$$

$$\therefore \text{gradient of normal} = -\frac{1}{p}$$

Equation of normal at P:

$$y - ap^2 = -\frac{1}{p}(x - 2ap) \quad \checkmark$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3 \quad \text{--- (1)}$$

ii) Equation of normal at Q:

$$x + qy = 2aq + aq^3 \quad \text{--- (2)}$$

① - ②,

$$\begin{aligned} y &= \frac{2a(p-q) + a(p^3 - q^3)}{p-q} \\ &= \frac{2a(p-q) + a(p-q)(p^2 + pq + q^2)}{p-q} \\ &= a(2 + p^2 + pq + q^2) \quad \checkmark \end{aligned}$$

Sub into ①,

$$\begin{aligned} x &= 2ap + ap^3 - ap(2 + p^2 + pq + q^2) \\ &= -apq(p+q) \quad \checkmark \end{aligned}$$

$$\therefore R = [-apq(p+q), a(2 + p^2 + pq + q^2)] \quad \text{--- (2)}$$

$$\text{iii) } x = -apq(p+q) \quad \text{--- (1)}$$

$$y = a(2 + p^2 + pq + q^2) \quad \text{--- (2)}$$

$$pq = -2 \quad \text{--- (3)}$$

Sub ③ into ①,

$$x = -a(-2)(p+q)$$

$$p+q = \frac{x}{2a} \quad \text{--- (4)} \quad \checkmark$$

From ②,

$$\begin{aligned} y &= a[pq + (p+q)^2 - 2pq + 2] \\ &= a[-2 + (\frac{x}{2a})^2 - 2(-2) + 2] \quad \text{using ③ \& ④} \\ &= \frac{x^2}{4a} + 4a \end{aligned}$$

$$\therefore x^2 = 4a(y - 4a) \quad \checkmark$$

is the locus of R

①

②

$$d) \int_0^1 x^2 \sqrt{1+3x^2} dx$$

$$u = 1+3x^2$$

$$\frac{du}{dx} = 6x$$

$$= \frac{1}{6} \int_1^4 \sqrt{u} du \quad \checkmark$$

$$= \frac{1}{6} \times \frac{2}{3} [u^{\frac{3}{2}}]_1^4$$

$$= \frac{2}{27} (8-1)$$

$$= \frac{14}{27} \quad \checkmark$$

②

$$e) i) N = 100 + Ae^{-0.09t} \quad \text{--- (1)}$$

$$\frac{dN}{dt} = -0.09Ae^{-0.09t} \quad \checkmark$$

$$= -0.09(N-100) \quad \text{using ①} \quad \text{--- (1)}$$

$$\text{ii) } N = 100 + Ae^{-0.09t}$$

$$\text{When } t=0, N=400$$

$$400 = 100 + A$$

$$\therefore A = 300$$

$$N = 100 + 300e^{-0.09t}$$

$$150 = 100 + 300e^{-0.09t} \quad \checkmark$$

$$\frac{50}{300} = e^{-0.09t}$$

$$\ln 6^{-1} = -0.09t$$

$$t = \frac{\ln 6}{0.09}$$

$$\approx 19.9$$

It takes 20 years for the number to reach 150. ②

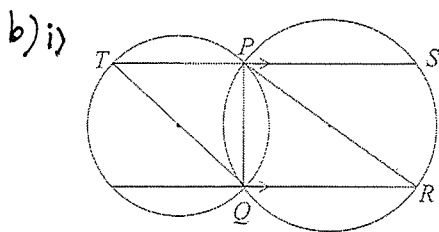
13) a) i) $V = \pi r^2 l$
 $d^2 = (2r)^2 + \left(\frac{l}{2}\right)^2$
 $= 4r^2 + \frac{l^2}{4}$
 $r^2 = \frac{1}{4} \left(d^2 - \frac{l^2}{4}\right)$
 $\therefore V = \pi l \left[\frac{1}{4} \left(d^2 - \frac{l^2}{4}\right)\right]$
 $= \frac{\pi l}{4} \left(d^2 - \frac{l^2}{4}\right)$

ii) $V = \frac{\pi l d^2}{4} - \frac{\pi l^3}{16}$
 $\frac{dV}{dl} = \frac{\pi d^2}{4} - \frac{3\pi l^2}{16}$
 $= 0$

When $\frac{\pi d^2}{4} = \frac{3\pi l^2}{16}$
 $l^2 = \frac{\pi d^2}{4} \times \frac{16}{3\pi}$
 $= \frac{4d^2}{3}$
 $l = \frac{2}{\sqrt{3}} d$

$\frac{d^2 V}{dl^2} = -\frac{3\pi l}{8}$
 $= -\frac{3\pi}{8} \times \frac{2d}{\sqrt{3}}$
 < 0

$\therefore \text{Max } V \text{ when } l = \frac{2}{\sqrt{3}} d$



$\angle PQR = 90^\circ$ (Angle in a semicircle) ✓

$\angle SPQ + \angle PQR = 180^\circ$ (Co-interior angles, $PS \parallel QR$)
 $\therefore \angle SPQ = 90^\circ$

Hence QS is a diameter (Angle in a semicircle equals 90°)

ii) $\angle TPQ = \angle PQR$ (Alternate angles, $TS \parallel QR$)
 $= 90^\circ$

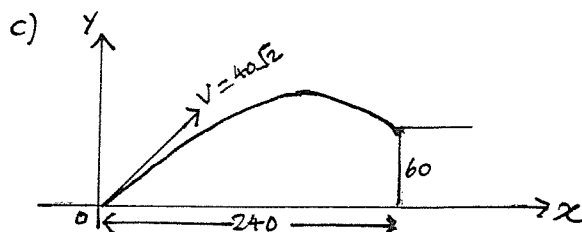
$\therefore TQ$ is a diameter (Angle in a semicircle = 90°) ✓

$PRQT$ is a parallelogram ($TQ \parallel PR$, $TP \parallel QR$)

$PR = QT$ (Opposite sides of parallelogram)

PR and QT are equal diameters ✓

$\therefore$ the two circles have equal radii.



i) $\ddot{y} = -10$

$\dot{y} = -10t + C_1$

When $t=0$, $\dot{y} = 40\sqrt{2} \sin 45^\circ$
 $= 40$

$\therefore C_1 = 40$

$\dot{y} = -10t + 40$

$y = -\frac{10t^2}{2} + 40t + C_2$

When $t=0$, $y=0$

$\therefore C_2 = 0$

$y = -5t^2 + 40t$ ✓

ii) $x = 40t$ — ①

$y = -5t^2 + 40t$ — ②

Sub ① into ②,

$y = -5\left(\frac{x}{40}\right)^2 + 40\left(\frac{x}{40}\right)$ ✓

$y = x - \frac{x^2}{320}$

iii) Sub $y=60$ into ②,

$60 = -5t^2 + 40t$ ✓

$t^2 - 8t + 12 = 0$

$(t-6)(t-2) = 0$

$t = 2 \text{ or } 6$

$\therefore$ it takes 6s to reach the top. ✓

iv) At the top of the cliff,

$\dot{x} = 40$

$\dot{y} = -10t + 40$

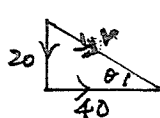
$= -10(6) + 40$

$= -20$ ✓

$V^2 = (-20)^2 + 40^2$

$= 2000$

$|V| = 20\sqrt{5}$ ✓



$\tan \theta = \frac{20}{40}$

$\theta = 26^\circ 34'$ ✓

The velocity is $20\sqrt{5} \text{ m/s}$,
at an angle of $26^\circ 34'$ to the horizontal.

14) a) $n=1$,
 $3^n + 7^{n+1} = 3 + 7^2$
 $= 52$ which is divisible by 4

$\therefore$ true for $n=1$

Assume true for $n=k$,

i.e. $3^k + 7^{k+1} = 4P$ where P is an integer

Prove true for $n=k+1$,

i.e. $3^{k+1} + 7^{k+2} = 4Q$, Q an integer

$$\begin{aligned} \text{LHS} &= 3 \times 3^k + 7^k \times 49 \\ &= 3(4P - 7 \times 7^k) + 49 \times 7^k \quad \text{using assumption} \\ &= 3 \times 4P - 21 \times 7^k + 49 \times 7^k \\ &= 3 \times 4P + 28 \times 7^k \\ &= 4(3P + 7 \times 7^k) \\ &= 4Q \end{aligned}$$

Hence result is true for $n=k$, it is also true for $n=k+1$.

By principal of mathematical induction, result is true for all integers $n \geq 1$. ③

b) i) $\ddot{x} = \frac{d}{dx} \left(\frac{v^2}{2} \right)$
 $= \frac{d}{dx} \frac{1}{2} (10-x)^2$
 $= \frac{d}{dx} (50 - 10x + \frac{1}{2}x^2)$ ✓
 $= x - 10$ ①

ii) $\frac{dx}{dt} = 10 - x$
 $\int \frac{1}{10-x} dx = \int dt$
 $t = -\ln(10-x) + C$ ✓
 When $t=0$, $x=0$
 $0 = -\ln 10 + C$
 $\therefore t = -\ln(10-x) + \ln 10$
 $= \ln \frac{10}{10-x}$ ✓
 $e^t = \frac{10}{10-x}$
 $10-x = 10 e^{-t}$

$x = 10 - 10e^{-t}$ ✓ ③

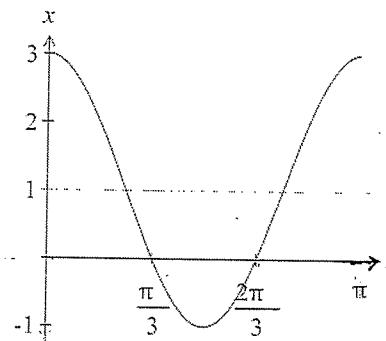
iii) 10 m on the RHS of O. ✓ ①

c) i) $x = 4 \cos^2 t - 1$
 $= 2 \cos 2t + 1$
 $\dot{x} = -4 \sin 2t$
 $\ddot{x} = -4 \times 2 \cos 2t$
 $= -4(x-1)$
 $= -2^2(x-1)$ ✓

$\therefore$ SHM about the point $x=1$ ✓ ②

ii) Period $= \frac{2\pi}{2}$
 $= \pi$ seconds ①

iii)



Shape ✓
 t intercepts ✓

iv) Velocity is increasing most rapidly when $\ddot{x}$ has the greatest positive value or x takes the least value, i.e. $x = -1$ ✓

$x = 4 \cos^2 t - 1$
 $-1 = 4 \cos^2 t - 1$
 $\cos^2 t = 0$
 $t = \frac{\pi}{2}$ ✓

OR From graph,
 when $x = -1$

$t = \frac{1}{2} \left(\frac{\pi}{3} + \frac{2\pi}{3} \right)$
 $= \frac{\pi}{2}$

$\therefore$ velocity is increasing most rapidly at $\frac{\pi}{2}$ seconds ②